LLMs Read Production Modes Better Than They Attribute Sources: A Controlled Study of Mode Detection and Latent Mode Structure

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Abstract

Large language models increasingly act as readers, judges, and filters for text, yet most detection work still asks only whether text is human- or machine-written. We study a broader question: can an LLM detect how a text was produced, and does that production mode behave like a latent variable in representation space? We build a controlled six-mode corpus from 60 HC3 anchors, balanced across open-domain QA, finance, and medicine, with matched variants for human originals, direct LLM generations, LLM-humanized text, human text polished by an LLM, dictated/spoken rewrites, and keyboard-noisy text. We then evaluate zero-shot mode classification with GPT-4.1, compare it to a grouped stylometric baseline, probe TEXT-EMBEDDING-3-LARGE embeddings with grouped cross-validation, and test whether mode-specific embedding deltas are consistent across content.

The results support a narrow but strong claim. On the controlled benchmark, GPT-4.1 achieves perfect six-way classification, while the stylometric baseline reaches 0.400 accuracy and the grouped embedding probe reaches 0.489 accuracy on unseen content anchors. All five transformed modes also show statistically significant within-mode delta consistency, with the strongest effects for keyboard noise and dictated speech. However, the same prompting setup reaches only 0.256 accuracy on six-way source attribution in SEMEVAL-2024 TASK 8 Subtask B and collapses most non-human systems into CHATGPT or human. We conclude that current LLMs read broad production modes well, but zero-shot generator identity remains substantially harder.

1 Introduction

Large language models do not only read *what* a text says. They also appear to infer *how* it was produced. That distinction matters because LLMs now sit inside moderation pipelines, educational-integrity tools, retrieval systems, and model-based evaluators. If those systems react differently to dictated text, edited text, or machine-written text, then production mode becomes part of the effective input to downstream decisions.

Why is this problem still open? Existing work already shows that machine-generated text detection is feasible in many settings Wu et al. [2023], that exact generator attribution can be benchmarked at scale Wang et al. [2024], and that process labels such as machine-humanized and human-machine-polished text are learnable Abassy et al. [2024]. Other lines of work isolate spoken-to-written conversion Liu et al. [2024] or behavioral traces such as keystroke dynamics Kundu et al. [2024]. What remains missing is a unified, content-controlled test of whether these tasks reflect a broader ability to read *production mode* rather than a collection of unrelated labels.

Our main question is simple. What text-production modes can current LLMs detect beyond binary human-vs-LLM authorship, and can mode be isolated as a latent variable in LLM-derived representations? We answer this question with a matched six-mode benchmark derived from HC3 Guo et al.

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[2023]. Each content anchor is realized in six forms: human original, direct LLM generation, LLM-humanized text, human text polished by an LLM, dictated/spoken rewriting, and keyboard-noisy text. We then test the problem from three angles: zero-shot LLM classification, a grouped stylometric baseline, and embedding-space analyses that ask whether mode generalizes across unseen content.

The headline result is asymmetric. On the controlled benchmark, GPT-4.1 classifies all six modes perfectly, while the stylometric baseline reaches 0.400 accuracy and the grouped embedding probe reaches 0.489 accuracy, well above the 0.167 chance rate. At the same time, the very same prompting setup reaches only 0.256 accuracy on six-way source attribution in SEMEVAL-2024 TASK 8 Subtask B. This split matters more than the perfect in-domain score. It suggests that LLMs are especially good at reading broad process states such as spoken cadence, keyboard noise, and post-editing, but much weaker at zero-shot fine-grained generator identity.

We make four contributions.

- We construct a controlled six-mode corpus with 360 texts from 60 matched HC3 anchors across three domains.
- We show that a real LLM reader, GPT-4.1, sharply separates broad production modes and substantially outperforms a grouped stylometric baseline.
- We provide latent-variable evidence for mode using grouped embedding probes and within-mode transformation-vector consistency tests.
- We bound the claim with an external negative result: zero-shot source attribution on SEMEVAL-2024 TASK 8 remains weak and collapses most non-human systems into a small subset of labels.

Paper organization. Section 2 positions our study relative to prior detection, attribution, and process-label work. Section 3 describes the corpus construction and evaluation protocol. Section 4 presents the main results, and Section 5 interprets their scope and limits before section 6 concludes.

2 Related Work

Binary detection and source attribution. Surveys of machine-generated text detection organize the space into statistical, neural, watermarking, and human-assisted methods, and emphasize that robustness remains unresolved Wu et al. [2023]. SEMEVAL-2024 TASK 8 established the strongest benchmark in our review for moving beyond binary labeling, with subtasks for human-vs-machine detection, exact generator attribution, and human-machine boundary detection Wang et al. [2024]. Our work builds directly on that framing, but we ask a different question: instead of starting from generator identity, we start from broad production modes and ask whether an LLM reader can recover them under approximate content control.

Fine-grained process labels. LLM-DETECTAIVE shows that machine-generated, machine-humanized, and human-machine-polished texts can be separated in a four-way setup Abassy et al. [2024]. Ji et al. argue that ambiguity itself deserves explicit treatment, since many texts do not fit cleanly into binary human-vs-AI labels Ji et al. [2024]. TEXTMACHINA complements this line by providing infrastructure for generating attribution and mixed-authorship datasets when public benchmarks are insufficient Sarvazyan et al. [2024]. We share the process-label motivation of these papers, but unlike them we keep content anchors roughly matched across multiple modes and pair end-task classification with embedding-space tests.

Spoken and behavioral signals. Not all production traces are tied to final-text authorship. SWAB frames spoken-to-written conversion as a document-level transformation problem and shows that transcript artifacts form a coherent mode that LLMs can rewrite Liu et al. [2024]. Kundu et al. show that AI assistance can also be detected from keystroke dynamics rather than final text alone Kundu et al. [2024]. These papers motivate two of our mode families: dictated/spoken rewrites and keyboard-noisy text. Our study focuses on text-only evidence, so it should be read as complementary to richer behavioral detection setups rather than a replacement for them.

Human- and model-authored comparison corpora. HC3 provides a practical source of matched human and ChatGPT answers across multiple QA domains Guo et al. [2023]. We use HC3 as the anchor corpus for our controlled benchmark because it gives us real human answers that can be transformed into multiple process variants. This lets us vary production mode while holding topic and intent approximately fixed, which is the key design element missing from prior work.

3 Methodology

Study design. We test whether production mode is both directly readable and latently represented. The first claim concerns observable performance: can an LLM classify how a text was produced? The second claim concerns representation: do embeddings encode reusable mode information that transfers across unseen content? We therefore combine direct zero-shot classification with grouped probing and transformation-vector analysis.

3.1 Controlled six-mode corpus

We sample 60 content anchors from HC3, balanced across `open_qa`, `finance`, and `medicine` with 20 anchors per domain. Each anchor contributes six variants, for a total of 360 texts with no missing values:

- `human_original`: the original human answer from HC3.
- `llm_generated`: a direct answer generated by GPT-4.1.
- `llm_humanized`: the LLM answer rewritten to appear more human-authored.
- `human_polished`: the human answer polished by GPT-4.1.
- `dictated_spoken`: the human answer rewritten into dictated or spoken-style prose.
- `keyboard_noisy`: a deterministic perturbation of the human answer using adjacent-key substitutions, transpositions, and omissions.

The dataset is intentionally matched by content anchor rather than by exact sentence preservation. This keeps the semantic target roughly fixed while allowing realistic mode changes. Word counts remain broadly comparable across modes, with mean lengths ranging from 95.2 words for `llm_humanized` to 111.6 words for `human_original`.

3.2 Models, baselines, and evaluation tasks

All API-backed generation and classification use GPT-4.1 at temperature 0. Embeddings come from TEXT-EMBEDDING-3-LARGE. We evaluate three complementary systems on the controlled corpus:

- **Zero-shot LLM judge.** The model predicts one of the six mode labels from text alone and returns calibrated probabilities.
- **Stylometric baseline.** We train a grouped baseline over hand-crafted features including length, punctuation rate, digit ratio, lexical diversity, typo-like token rate, disfluency markers, and sentence statistics.
- **Embedding probe.** We train a one-vs-rest logistic regression classifier on TEXT-EMBEDDING-3-LARGE vectors after `StandardScaler` and PCA retaining 95% of variance.

We use 5-fold `GroupKFold` by content anchor for the stylometric baseline and the embedding probe. This grouped protocol prevents lexical leakage from near-duplicate mode variants of the same anchor from appearing in both train and test folds.

3.3 Latent-mode analysis

We test the latent-variable hypothesis in two ways. First, we evaluate whether the grouped embedding probe predicts mode on unseen content above the 1/6 chance rate. Second, for each transformed mode, we compute the embedding delta from `human_original` to the transformed variant and compare within-mode cosine similarities against between-mode baselines. If mode behaves like a reusable representational direction, within-mode deltas should align more than cross-mode deltas.

3.4 External validation

To test whether the observed effect transfers beyond our custom corpus, we evaluate zero-shot source attribution on SEMEVAL-2024 TASK 8 Subtask B Wang et al. [2024]. We sample 30 development examples from each of six classes: human, ChatGPT, Cohere, Davinci, Bloomz, and Dolly, for

Table 1: Controlled six-mode classification results. Higher is better for all metrics. Best results are in **bold**.

Method	Accuracy	Macro-F1
GPT-4.1 zero-shot judge	**1.000**	**1.000**
Stylometric baseline	0.400	0.386
Embedding probe	0.489	0.477

Table 2: Per-class recall for the grouped embedding probe on unseen content anchors. Overt production shifts are easiest to recover from embeddings alone.

Mode	Recall
`dictated_spoken`	**0.700**
`llm_generated`	0.617
`keyboard_noisy`	0.567
`llm_humanized`	0.567
`human_original`	0.250
`human_polished`	0.233

180 texts total. This task is deliberately harder than the controlled corpus because it asks for exact generator identity rather than broad process state.

3.5 Metrics, statistics, and implementation details

We report accuracy and macro-F1 for all multiclass tasks, plus confusion matrices to expose near-miss structure. For the controlled corpus, we estimate 95% bootstrap confidence intervals for the LLM judge and use paired bootstrap comparisons against the stylometric baseline. For the embedding probe, we run a corrected 20-run permutation test against shuffled-label nulls. For transformation-vector consistency, we compare within-mode and between-mode cosine similarities with per-mode tests and apply Benjamini-Hochberg correction across modes.

The pipeline runs in Python 3.12.8 with `openai 2.38.0`, `scikit-learn 1.8.0`, `scipy 1.17.1`, `pandas 3.0.3`, `matplotlib 3.10.9`, and `seaborn 0.13.2`. GPUs were available but not used in the final pipeline because all learned components were lightweight CPU-based models and all LLM stages were API-backed. Recorded non-embedding usage totals 227,642 tokens across generation, custom-mode classification, and SemEval-2024 Task 8 classification.

4 Results

Main benchmark results. Table 1 compares direct LLM judgment, the grouped stylometric baseline, and the grouped embedding probe on the controlled six-mode corpus. GPT-4.1 reaches perfect accuracy and macro-F1, while the stylometric baseline reaches 0.400 accuracy and the embedding probe reaches 0.489. The paired bootstrap gap between the LLM judge and the stylometric baseline is +0.602 accuracy (95% CI: [0.555, 0.650]) and +0.618 macro-F1 (95% CI: [0.573, 0.665]).

The task is easy, but not vacuous. Figure 1 shows a perfectly diagonal confusion matrix for the controlled benchmark. The model does not appear to solve the task only by assigning near-certain probabilities to every example: representative correct-class probabilities remain moderate, including 0.75 for `human_polished`, 0.78 for `dictated_spoken`, and 0.80 for `llm_humanized`. That pattern is consistent with a highly separable task rather than a degenerate prompt that simply elicits one fixed label.

Embedding probes recover mode above chance. The grouped embedding probe reaches 0.489 accuracy on unseen content anchors, well above the 0.167 chance rate. A corrected 20-run permutation test yields shuffled accuracies between 0.139 and 0.203 and an observed $p = 0.0476$. Table 2 shows that the probe separates overt mode shifts most clearly: `dictated_spoken` reaches 0.700 recall, while the subtler `human_original` and `human_polished` classes remain much harder.

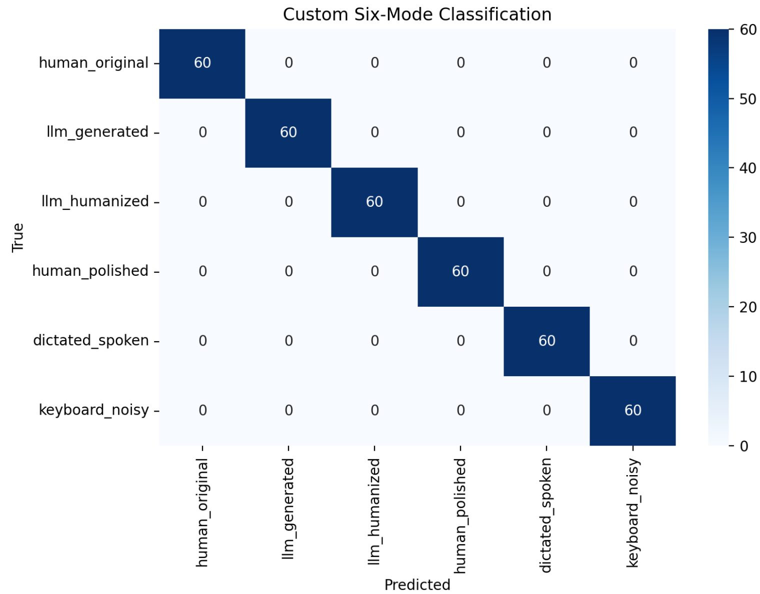

Figure 1: Confusion matrix for zero-shot six-way mode classification on the controlled corpus. The matrix is perfectly diagonal, but the underlying probability outputs are not saturated: representative correct-class probabilities range from 0.75 to 0.94.

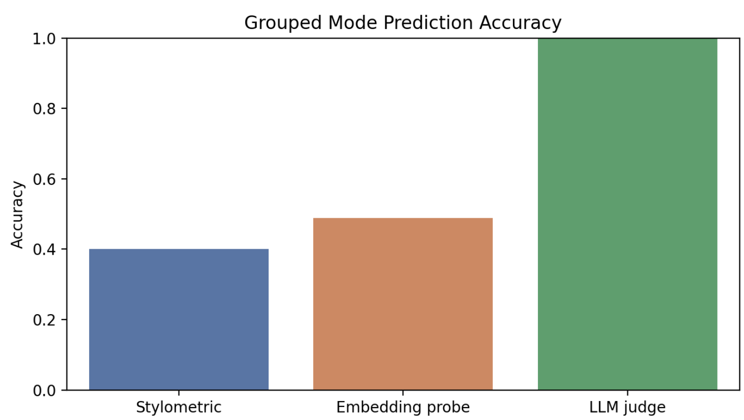

Figure 2: Grouped embedding-probe accuracy against the shuffled-label null distribution. The observed score of 0.489 sits well above the null range, which supports the claim that mode information survives content-grouped evaluation.

Mode deltas are directionally consistent. Table 3 summarizes the transformation-vector analysis. All five transformed modes show higher within-mode cosine similarity than between-mode similarity after Benjamini-Hochberg correction. The largest effects occur for `keyboard_noisy` and `dictated_spoken`, while `human_polished` is statistically significant but much weaker.

External validation sharply narrows the claim. Strong in-domain mode recognition does not translate into robust zero-shot source attribution. Table 4 shows that GPT-4.1 reaches only 0.256 accuracy and 0.136 macro-F1 on SemEval-2024 Task 8 Subtask B, only modestly above the 0.167 chance rate. The bootstrap 95% confidence interval for accuracy is [0.194, 0.311]. Figure 4 shows why: the model correctly identifies 29/30 human examples and 17/30 ChatGPT examples, but it collapses almost all Cohere, Davinci, Bloomz, and Dolly examples into the human or ChatGPT labels.

Table 3: Transformation-vector consistency for each mode. We compare cosine similarity among deltas from `human_original` to a transformed variant. Higher within-mode than between-mode similarity supports a reusable mode direction.

Mode	Within mean	Between mean	BH-adjusted p
`keyboard_noisy`	**0.111**	-0.015	$< 10^{-121}$
`dictated_spoken`	0.094	0.027	$< 10^{-60}$
`llm_humanized`	0.058	0.034	4.13×10^{-13}
`llm_generated`	0.036	0.014	6.67×10^{-9}
`human_polished`	0.029	0.021	6.40×10^{-3}

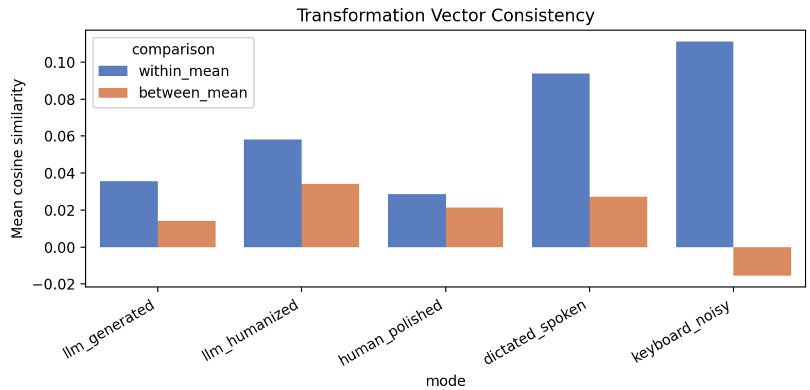

Figure 3: Within-mode versus between-mode cosine similarity for embedding deltas from `human_original` to transformed variants. Larger gaps indicate more reusable mode-specific directions. Keyboard noise and dictated speech produce the clearest latent shifts.

What pattern emerges across analyses? The results consistently point to a hierarchy of detectable modes. Overt production changes such as keyboard noise and spoken-style dictation are strongest in both direct classification and latent-space consistency. Post-editing modes remain detectable, but they are weaker both in per-class probe recall and in delta alignment. Exact zero-shot source identity is weaker still. This ordering is the core empirical claim of the paper.

5 Discussion

What do the results show? The main finding is not that LLMs can fingerprint arbitrary generators. The more defensible claim is narrower: current LLMs can read broad production modes from text alone, and that signal is strong enough to appear both in direct judgments and in embedding-space analyses. The controlled benchmark, grouped probe, and delta-consistency tests all support that view. The external SEMEVAL-2024 TASK 8 result then provides the necessary boundary condition by showing that zero-shot generator identity is much less stable.

Why does the split matter? If a model can separate dictated text, keyboard-noisy text, and LLM-post-edited prose, then mode already affects systems that use LLMs as judges or filters. Provenance screening, moderation triage, and educational-integrity workflows may therefore be reacting to more than semantic content. At the same time, our external result argues against overinterpreting these abilities as reliable model fingerprinting. In practice, a detector that is good at reading process state may still be poor at naming the exact generator family.

The latent-variable evidence is promising but partial. The grouped probe and the delta-similarity tests support the idea that mode behaves like a reusable direction in representation space. That matters because it suggests mode is not merely a prompt artifact attached to one benchmark. Still, the evidence is not yet definitive. The perfect custom classification score indicates that the benchmark is relatively easy, and same-family generation and classification may amplify the signal.

Table 4: External zero-shot source attribution on SemEval-2024 Task 8 Subtask B. Chance accuracy is 0.167. The confidence intervals come from bootstrap resampling.

Metric	Score	95% CI
Accuracy	0.256	[0.194, 0.311]
Macro-F1	0.136	[0.108, 0.164]

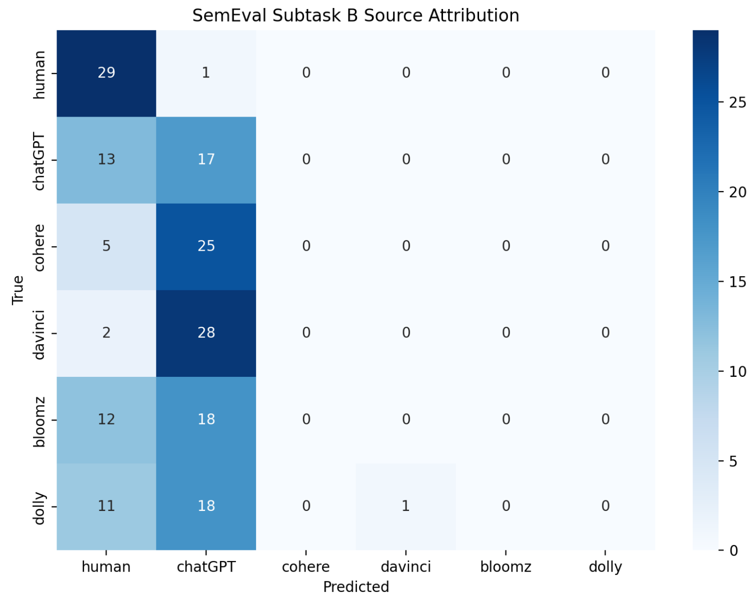

Figure 4: Confusion matrix for zero-shot six-way source attribution on SemEval-2024 Task 8 Subtask B. Most non-human systems collapse into ChatGPT or human, which indicates that broad process cues transfer more readily than exact generator identity.

Limitations. Several threats to validity remain. First, GPT-4.1 generated several transformed modes and also served as the judge, which raises a self-recognition risk. Second, the `keyboard_noisy` mode is synthetic and deterministic rather than collected from real typing data. Third, content is only approximately matched across variants, so some semantic drift may remain. Fourth, the permutation test is exploratory because it uses only 20 corrected runs. Fifth, the SemEval-2024 Task 8 experiment is zero-shot only; a tuned attribution model would almost certainly perform much better. Finally, our analysis is document-level and does not test span-level mode transitions such as those in SemEval-2024 Task 8 Subtask C.

Broader implications. These results support a more cautious vocabulary for provenance claims. Broad production mode is likely a real and useful construct for ambiguity-aware evaluation and screening, but exact source attribution should be treated as a separate and harder problem. We expect the strongest near-term applications to involve abstaining or flagging mode uncertainty rather than making high-confidence claims about exact model identity.

6 Conclusion

We study whether LLMs detect production modes beyond binary authorship and whether those modes behave like latent variables in representation space. Using a controlled six-mode corpus with matched HC3 anchors, we find that GPT-4.1 perfectly separates broad production modes, substantially outperforms a grouped stylometric baseline, and leaves measurable mode structure in text-embedding-3-large embeddings. At the same time, the same prompting setup performs poorly on external six-way source attribution in SemEval-2024 Task 8 Subtask B.

The key takeaway is straightforward: LLMs read broad process modes well, but zero-shot generator identity remains much harder.

The next step is to make this claim harder to satisfy. Future work should decouple generator and judge model families, replace synthetic proxies with real dictated-versus-typed and keyboard-derived data, add abstention and ambiguity labels for borderline cases, and test whether the learned mode directions transfer to mixed-authorship benchmarks such as SEMEVAL-2024 TASK 8 Subtask C.

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